

EXPERIMENT – 2

Title

To Create a Table, Alter and Drop Table

Aim / Objective

To understand and implement SQL Data Definition Language (DDL) commands by creating a table, modifying its structure using the **ALTER TABLE** command, and removing it using the **DROP TABLE** command.

Theory

Data Definition Language (DDL) is a subset of SQL used to define and manage the structure of database objects such as databases, tables, indexes, and views.

The commonly used DDL commands are:

- **CREATE** – Creates database objects.
- **ALTER** – Modifies the structure of existing database objects.
- **DROP** – Deletes database objects permanently.
- **TRUNCATE** – Removes all records from a table while keeping its structure.
- **RENAME** – Changes the name of a database object.

In this experiment, students will learn how to create a table, modify its structure by adding, modifying, renaming, and dropping columns, and finally remove the table from the database.

Learning Objectives

After completing this experiment, students will be able to:

- Understand DDL commands.
 - Create database tables.
 - Modify the structure of tables.
 - Add and remove columns.
 - Rename columns.
 - Delete tables from the database.
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Software Requirements

- Oracle Database
 - Oracle SQL*Plus / Oracle SQL Developer
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Algorithm / Procedure

1. Open Oracle SQL Developer or SQL*Plus.
 2. Connect to the Oracle database.
 3. Create a table.
 4. Insert sample records.
 5. Alter the table structure.
 6. Display the modified table structure.
 7. Drop the table.
 8. Verify that the table has been removed.
 9. Stop.
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Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee  
(  
  EmpID NUMBER(5),  
  EmpName VARCHAR2(30),  
  Department VARCHAR2(30),  
  Salary NUMBER(10,2)  
);
```

Step 2 : Display Table Structure

```
DESC Employee;
```

Expected Output

Name	Null Type
EMPID	NUMBER(5)
EMPNAME	VARCHAR2(30)
DEPARTMENT	VARCHAR2(30)
SALARY	NUMBER(10,2)

Step 3 : Insert Sample Records

```
INSERT INTO Employee  
VALUES  
(101,'Rahul','IT',35000);
```

```
INSERT INTO Employee  
VALUES  
(102,'Priya','HR',42000);
```

```
COMMIT;
```

Step 4 : Display Table

```
SELECT * FROM Employee;
```

Expected Output

EmpID	EmpName	Department	Salary
101	Rahul	IT	35000
102	Priya	HR	42000

ALTER TABLE Operations

Program 5 : Add a New Column

```
ALTER TABLE Employee
```

```
ADD Email VARCHAR2(50);
```

Verify

DESC Employee;

Expected Output

Name	Type
EmpID	NUMBER
EmpName	VARCHAR2(30)
Department	VARCHAR2(30)
Salary	NUMBER
Email	VARCHAR2(50)

Program 6 : Modify Column Size

```
ALTER TABLE Employee  
MODIFY EmpName VARCHAR2(50);
```

Verify

DESC Employee;

Program 7 : Rename Column

```
ALTER TABLE Employee  
RENAME COLUMN Department TO DeptName;
```

Verify

DESC Employee;

Expected Output

Name

EmpID

EmpName

DeptName

Salary

Email

Program 8 : Drop a Column

```
ALTER TABLE Employee
```

```
DROP COLUMN Email;
```

Verify

```
DESC Employee;
```

Expected Output**Name**

EmpID

EmpName

DeptName

Salary

Program 9 : Rename Table

```
ALTER TABLE Employee
```

```
RENAME TO Employee_Details;
```

Verify

```
SELECT * FROM Employee_Details;
```

DROP TABLE

Program 10 : Drop Table

```
DROP TABLE Employee_Details;
```

Verify

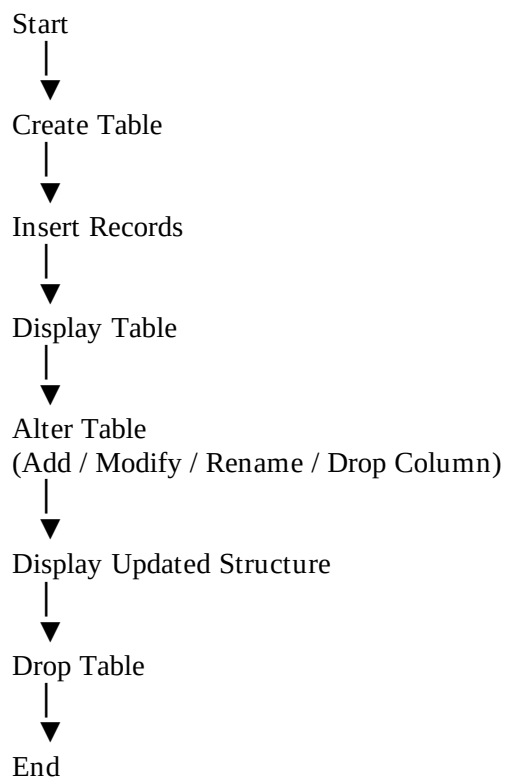
```
DESC Employee_Details;
```

Expected Output

ERROR:

Table or View does not exist.

Flowchart



Explanation

CREATE TABLE

Creates a new table in the database.

DESC

Displays the structure of the table.

ALTER TABLE

Used to modify the structure of an existing table.

ADD

Adds a new column.

MODIFY

Changes the datatype or size of a column.

RENAME COLUMN

Changes the name of a column.

DROP COLUMN

Removes a column permanently.

RENAME TO

Changes the table name.

DROP TABLE

Deletes the table permanently from the database.

Observation

- Table was created successfully.

- Sample records were inserted.
 - Table structure was modified using ALTER TABLE.
 - Columns were added, modified, renamed, and dropped successfully.
 - Table was renamed successfully.
 - Table was deleted using the DROP command.
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Result

The SQL DDL commands **CREATE TABLE**, **ALTER TABLE**, and **DROP TABLE** were successfully executed. The experiment demonstrated how to create a table, modify its structure, and permanently remove it from the database.

Applications

- Employee Management Systems
 - Banking Systems
 - Hospital Management Systems
 - College Management Systems
 - Inventory Management Systems
 - Railway Reservation Systems
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Advantages

- Easy creation of database tables.
 - Flexible modification of table structure.
 - Efficient database design.
 - Supports schema evolution.
 - Simplifies database administration.
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Viva Questions

1. What is DDL?
2. What is the purpose of the CREATE TABLE command?

3. What is the ALTER TABLE command used for?
 4. What is the difference between MODIFY and ADD?
 5. What is the purpose of DROP TABLE?
 6. What is the difference between DROP TABLE and TRUNCATE TABLE?
 7. What does the DESC command display?
 8. Can a dropped table be recovered?
 9. What is the purpose of RENAME TO?
 10. Why are DDL commands important?
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Lab Assignment

1. Create a **Student** table with five columns.
 2. Add a new column named **Email**.
 3. Modify the size of the Student Name column.
 4. Rename the **Course** column to **Program**.
 5. Drop the **Email** column.
 6. Rename the Student table to **Student_Details**.
 7. Display the table structure after each modification.
 8. Drop the Student_Details table.
 9. Compare CREATE, ALTER, and DROP commands.
 10. Explain the importance of DDL commands with suitable examples.
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Precautions

- Verify the table name before executing the DROP TABLE command, as it permanently deletes the table.
 - Ensure that no important data is lost before altering or dropping database objects.
 - Execute DDL commands one at a time and verify the results using the DESC command.
 - Use meaningful names for tables and columns.
 - Commit transactions where required after data insertion.
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Learning Outcome

After completing this experiment, students will be able to:

- Create database tables using the CREATE TABLE command.
- Modify table structures using various ALTER TABLE operations.
- Add, modify, rename, and delete columns.
- Rename and remove tables from the database.
- Apply DDL commands effectively while designing and managing relational databases.